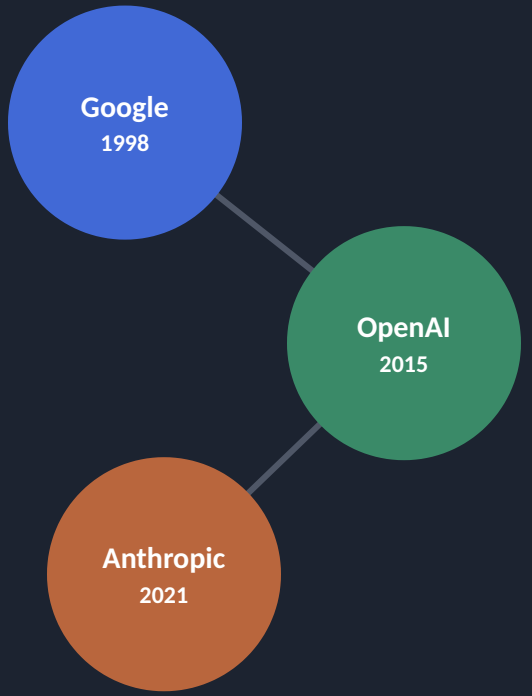
PICUP PITTSBURGH 2026

Agentic AI Overview



Adapted from slides by Larry Engelhardt

Three companies at the center of today's AI

1998 Google

- Dominant search engine, then ads, Android, YouTube, cloud
- Acquired DeepMind (founded 2010) in 2014

Search → DeepMind → Gemini

2015 OpenAI

- Began as a nonprofit lab aiming at AGI for broad human benefit
- For-profit arm in 2019; a public benefit corporation since 2025. Microsoft a major investor

GPT → ChatGPT → Codex

2021 Anthropic

- Left OpenAI amid disagreements over direction and safety
- Amazon and Google are major investors

Claude → Claude Code

Two breakthroughs in the same five years

AI for scientific discovery

DeepMind (Google)

2016 AlphaGo's "Move 37": a strategy human experts hadn't considered

2020 AlphaFold 2 cracks the ~50-year protein-folding problem

2024 Chemistry Nobel for predicting protein structure with AlphaFold (Hassabis & Jumper, shared with David Baker for protein design)

Large language models

Google → OpenAI

2017 Google's "Attention Is All You Need" introduces the Transformer

2018+ OpenAI scales it up: GPT-1 → GPT-2 → GPT-3 → GPT-4

2022 ChatGPT gives everyone a general-purpose language interface

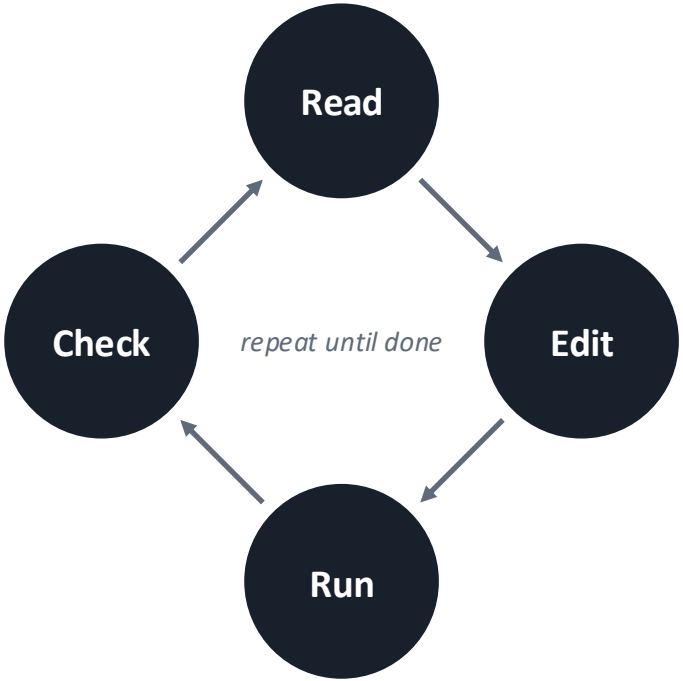
For physicists: the 2024 Nobel Prize in Physics went to John Hopfield and Geoffrey Hinton (Sutskever's PhD advisor) for bringing spin-system physics and statistical mechanics to neural networks: the Hopfield network and the Boltzmann machine, precursors of today's deep learning.

Today: AI agents that write and run code

A chatbot answers questions. An agentic coding tool does the work:

- Reads the whole project
- Edits files and runs code in a terminal
- Tests its work, finds mistakes, tries again
- Splits big jobs among sub-agents

"Give the AI a computer and let it work on the project."



The big three: platforms and models

Company:	Anthropic	OpenAI	Google
Platform for Agentic Programming	Claude Code (Claude)	Codex (ChatGPT)	Antigravity (Gemini)
Models	Fable 5.1	Astra GPT-6	
	Opus 5.5	Sol GPT-6.1	Gemini 4 Argon
	Sonnet 5.5	Terra GPT-6.1	
	Haiku 5.5	Luna GPT-6	Gemini 3.7 Flash

Each company pairs an agentic coding tool with a family of models, from large to small.

Models as of November 2026

The big three: platforms and models

Company:	Anthropic	OpenAI	Google	Smartness
Platform for Agentic Programming	Claude Code (Claude)	Codex (ChatGPT)	Antigravity (Gemini)	
Models	Fable 5.1	Astra GPT-6		**Whoa!**
	Opus 5.5	Sol GPT-6.1	Gemini 4 Argon	Very Good
	Sonnet 5.5	Terra GPT-6.1		Pretty Good
	Haiku 5.5	Luna GPT-6	Gemini 3.7 Flash	Fast & cheap

Ways to use Claude Code or Codex

1. **Web browser**
   Agent works on a cloud copy of your GitHub repo
2. **CLI (Command Line Interface)**
   Do this if you like command lines!
3. **Desktop App**
   More user friendly than CLI
4. **VS Code Extension**
   Nice balance between CLI & Desktop App

Models as of November 2026 · Plain chat answers questions; it doesn't run your project

Beyond the big three

Other agentic coding tools worth knowing

META

Muse Code

New (beta); built for long-running, multi-agent coding

AMAZON / AWS

Kiro

Agentic IDE and command line with spec-driven development

GITHUB / MICROSOFT

Copilot

Now an agent that works on GitHub issues and pull requests

ANYSPHERE

Cursor

AI-first code editor whose agents can change a whole project

COGNITION

Devin

An "AI software engineer" in its own cloud workspace (also Windsurf)

OPEN SOURCE

Kilo Code

Choose from 500+ models, including xAI's Grok

KEY CHOICE

One company's agent + model
e.g. Claude Code with Claude

vs.

An agent where you pick the model
e.g. Kilo Code or OpenCode

A PERMANENT ASSESSMENT SHIFT IN PHYSICS

The artifact no longer certifies the learner

AI has broken a familiar inference: polished work no longer provides sufficient evidence of individual understanding.

THE OLD INFERENCE

Polished artifact → The learner understands

AI-ERA REALITY

Polished artifact → What a learner-AI system can produce

ASSESS VALIDATION AUTHORITY

Can the learner establish why this computational claim should be trusted?

A RESPONSE

Add a supervised defense lasting 5 to 10 minutes to one AI-vulnerable assignment.

Explain the model • Identify delegation • Justify the checks • Interpret the result • State the limits

Larry Engelhardt

Full framework: arxiv.org/abs/2607.15518

AI and Education

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https://aiandeducation.mit.edu/report/